

shane**scott**

algorithms, geometry, software, mathematics

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skills education

algorithm design and optimization
mathematical visualization
technical communication
clean code development
computational geometry
computer aided design
, C++ Python

Ph.D. in Mathematics

2018

Georgia Institute of Technology

thesis in surface topology, minor in quantum computation

Dual B.S. in Physics and Mathematics

2012

Kansas State University

thesis in wavelet analysis, GPA 4.0, year at University of Hyderabad, India

technical experience

Automating Engineering Design Optimization

2024-2026

Researcher at Braid Technologies

Led geometry team developing large-scale modern C++ computational geometry system and physics-informed modeling tools for next-generation engineering software. Algorithms for rapid, in-optimization-loop, multirepresentation geometry generation. Cross-functional work with optimization, simulation, and software engineering teams to enable AI guided geometry generation for structural engineering optimization applications. Responsible for geometry kernel architecture, technical road map, research strategy, and mentoring junior engineers. Delivered design optimization PoCs for major Japanese automotive and industrial OEMs

3D Printing Toolpath Control

Jan - Dec 2023

Software Engineer at UltiMaker

Developed control algorithms (slicer) for the Method line of FDM printers. Led projects in print time prediction and thermal behavior modeling for thermoplastics. Cross functional work from C++ build engineering, Python prototyping, and printer hardware.

Geometry-as-Keyword Search Engine

2021 - 2023

SDE3 Computation Geometry at Physna

Led R&D for geometric search and comparison for CAD and mesh models. Designed, implemented, and maintained a C++ geometry processing library for CAD tessellation and mesh analysis. Implemented and optimized shape retrieval algorithms for multi-million model geometric search index Thangs.

Automating Design and Print Preparation for Metal 3D Printing

2019 - 2021

Software Engineer at Divergent3D

Developed FEA geometry kernel for automating design via topology optimization, print segmentation, and print packing. Tech lead on optimization schemes for part segmentation and 3D printer packing. Collaboration on internal tools for structural engineers, CAD designers, and additive manufacturers.

Predicting Traffic Accidents from Driver GPS Data

2018 - 2019

Statistical Modeler at LexisNexis Risk

Physical models for GPS driver safety rating. Large data manipulation with high performance computing cluster. Rating driver risk and AI driver recognition using GPS data from many device types.

Calculating Surface Symmetries

2013 - 2018

Graduate Student Researcher at Georgia Institute of Technology

Research in abstract geometry and topology. Reconstruction problems in symmetries of surfaces. Algorithms in computing novel 3-manifold invariants using hyperbolic triangulation.

Vascular Geometry Segmentation for Bloodflow Simulation

June - Aug 2017

Computation Intern at Lawrence Livermore National Lab

Created geometry based computational load balancing for HARVEY, human blood flow simulation. Implemented algorithms automating decomposition of real scanned human vascular structure.

peer-reviewed publications

Combinatorial models for surface and free group symmetries. PhD diss., Georgia Institute of Technology, 2018. hdl.handle.net/1853/60722.

Exact computation of the n -loop invariants of knots. *Experimental Mathematics*. 25. 2 (2016). Garoufalidis, Sabo, and Scott.

Computing the partial word avoidability indices of ternary patterns. *Combinatorial Algorithms. IWOCA* (2012). *Lecture Notes in Computer Science*, vol 7643. Springer, Berlin, Heidelberg. Blanchet-Sadri, Lohr, and Scott.

Computing the partial word avoidability indices of binary patterns. *Journal of Discrete Mathematics* 23 (2013). Blanchet-Sadri, Lohr, and Scott.

Delay control in attosecond pump-probe experiments. *Optical Express* 17.24 (2009). Chini, Mashiko, Wang, Chen, Yun, Scott, Gilbertson, and Chang.

conference presentations

Topologists Outside Academia

2022, 2024

Topology Students Workshop, School of Mathematics, Georgia Institute of Technology

Presenting with Inkscape and Sozi

July 2016

Topology Students Workshop, School of Mathematics, Georgia Institute of Technology

Avoiding Patterns in Partial Words

July 2012

23rd International Workshop on Discrete Algorithms, Tamil Nadu, India

Ternary Patterns in Partial Words

April 2012

American Mathematical Society Spring Sectional Meeting, University of Kansas

teaching experience

Graduate Student Instructor

2012 - 2018

Georgia Institute of Technology

taught courses ranging from 20 to 120 students; managed teams of 2 to 5 teaching assistants; award winning instruction; subjects include calculus, differential equations, linear algebra, combinatorics, and algorithms

Georgia High School Mathematics Competition Organizer

2016 - 2017

Georgia Institute of Technology

coordinated annual statewide math competition of 400 students; managed team to design competition materials and activities; designed optical mark recognition automatic grading system

achievements

Access Ally Award

2017

Georgia Institute of Technology Office of Disability Services

awarded for impact on hearing-impaired student success, accessibility, and advocacy

Outstanding Graduate Teaching Assistant

2016

Georgia Institute of Technology School of Mathematics

chosen by the department to represent school for superior instruction

School of Mathematics Graduate Representative

2016 - 2017

Georgia Institute of Technology

represented graduate student body on the faculty graduate committee and the graduate student council; founding member of the graduate student chapter of the American Mathematical Society

Eagle Scout and Community Service Award

2007

Boy Scouts of America and Survivors of the Dodge City Mexican Village

awarded for the design and erection of a historical marker for the Mexican Village in Dodge City, KS